

Supplementary Information

LongPhase 2: uncertainty-aware co-phasing of small variants, structural variants and DNA methylation from long reads

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Supplementary Notes

Supplementary Note 1. Reading the phasing graph export

With `-dot`, `longphase phase` writes one file per chromosome, `<prefix>.<chrom>.dot`, containing one line per voting pair, `P.a -> Q.b [label=w]`, where `P` and `Q` are 1-based positions, `a`, `b` $\in \{1, 2\}$ index the reference (1) and alternate (2) allele, and `w` is the weighted haplotype vote accumulated at `P` for that allele at the time the vote was cast. The companion VCF carries, for every phased variant, the fields `INFO/H1`, `INFO/H2` (weighted votes for haplotype 1 and 2) and `INFO/PE` (their normalized Shannon entropy). The `gnn` sub-command reads both files; users who wish to inspect why a variant was unphased can therefore reconstruct its local window without re-running the phaser.

Supplementary Note 2. Reproducing the demonstration

The repository ships a 500-kb demonstration (`chr1:121.5–122.0 Mb`; `demo/demo.sh`) that runs `modcall`, SNV-only and four-class `phase`, `gnn` correction of both outputs and `compare` against a truth VCF, and prints switch errors, phased SNV counts and Hamming distance for the four configurations. Total runtime is under two minutes on four threads.

Supplementary Tables

Supplementary Table 1: Node features of the graph neural network. Each allele node of a window carries 31 features; # is the 1-based index in the feature vector (`GNNProcess.cpp`, `f[0]–f[30]`), listed here by group, not by index. h_1, h_2 : weighted haplotype votes (taken from the DOT edge labels when available, otherwise from INFO/H1, INFO/H2); PE: phasing entropy; PS: phase set; “centre” is the trigger variant of the window. Genomic-context features are computed from the reference when `-r` is given and are zero otherwise.

#	Feature	Definition
<i>Phasing evidence (16)</i>		
1	PE	normalized Shannon entropy of (h_1, h_2) , 0–1 bit
2	HP	haplotype of this allele node (genotype index)
3	gt_ref	genotype index of the reference allele (0)
4	gt_alt	genotype index of the alternate allele (1)
5	is_center	1 if the variant is the window trigger
6	is_same_ps	1 if the variant shares the trigger’s PS
7	allele	0 for the reference-allele node, 1 for the alternate-allele node
8	rel_offset	$(\text{pos} - \text{pos}_c) / \max \text{pos} - \text{pos}_c $ over the window, signed
9	H1	h_1
10	H2	h_2
11	total_votes	$h_1 + h_2$
12	vote_ratio	$\max(h_1, h_2) / (h_1 + h_2)$
13	log_total	$\log_2(1 + h_1 + h_2)$
14	log_h1	$\ln(1 + h_1)$
15	log_h2	$\ln(1 + h_2)$
25	rel_vote_depth	$\log_2((h_1 + h_2) / \overline{(h_1 + h_2)}_{\pm 10 \text{ variants}})$
<i>Genomic context (6)</i>		
16	hp_len	homopolymer run length at the variant (capped at 20)
17	gc_content	GC fraction in ± 50 bp
18	str_context	longest tandem-repeat unit count (unit 2–4 bp) in ± 15 bp, /10, capped at 1
19	seq_entropy	Shannon entropy of the ± 15 bp sequence (bits)
20	neigh_dens	$(\text{number of variants within } \pm 500 \text{ bp} - 1) / 20$
24	dist_near_indel	$\log_{10}(1 + \text{distance to nearest indel})$, capped at $\log_{10}(10001)$
<i>Graph structure (5)</i>		
21	out_degree	number of distinct target positions of outgoing DOT edges, /10, capped at 1
22	ps_block_size	$\ln(1 + \text{variants in the PS}) / \ln(101)$
23	log_center_dist	$\log_{10}(1 + \text{pos} - \text{pos}_c)$
26	out_weight_std	s.d. of outgoing DOT edge weights, /100, capped at 1
27	is_bridge	1 if removing the variant disconnects the window graph
<i>Variant class (4, one-hot)</i>		
28–31	is_SNV, is_indel, is_SV, is_5mC	variant class indicators

Supplementary Table 2: Edge features of the graph neural network. Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node’s incident edges. w is the DOT edge label (accumulated vote weight).

#	Feature	Definition
1	weight	$1/(1 + e^{-w})$
2	log_dist	$\log_{10}(1 + \text{pos}_{\text{dst}} - \text{pos}_{\text{src}})$
3	log_weight	$\ln(1 + w)$
4	is_same_hp	1 if the two allele nodes have the same allele index
5	is_inter_ps	1 if the two variants belong to different phase sets
6	weight_ratio	$w / (\text{sum of outgoing edge weights of the source allele node})$

Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.

Component	Setting
Input normalization	BatchNorm over 31 node features (frozen statistics at inference)
Input projection	Linear $31 \rightarrow 128$, GELU
Layers	4 GPS layers, hidden size 128
Local branch	GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \rightarrow 128$
Global branch	multi-head self-attention, 4 heads, Q/K/V/O projections $128 \rightarrow 128$
Fusion	$x \leftarrow x + \text{local} + \text{global}$; LayerNorm ($\varepsilon = 10^{-5}$)
Feed-forward	$128 \rightarrow 256 \rightarrow 128$, GELU, residual, LayerNorm
Classifier	concat $[h_i, \sum_j \sigma(w_{ij}), \max_j \sigma(w_{ij})]$ ($130 \rightarrow 128$ (GELU) $\rightarrow 2$, softmax; σ logistic, self-loop excluded)
Parameters	687,358 (float32, 2.75 MB, embedded in the binary)
Window	trigger PE ≥ 0.8 ; ± 20 variants (≤ 41 variants, ≤ 82 allele nodes; a 256-node cap exists but is never reached at this window size)
Decision	unphase if mean $P(\text{error}) \geq 0.30$ over the two allele nodes and all covering windows
Post-processing	split disconnected phase sets (largest component keeps PS); bridge protection off
Training	see Supplementary Table 4

Supplementary Table 4: Training configuration of the graph neural network. Values are those recorded in the released checkpoint `model_e6.pt` and in `train_e6.log`; every setting is the default of `train_gnn_29.py` except `-binary`, which was passed explicitly. “Scored nodes” are labelled allele nodes inside the central zone of a window, the only nodes that contribute to the loss.

Item	Setting
<i>Data</i>	
Source runs	HG002 ONT R10.4.1, four-class co-phasing, 10/12/14/16/18/20 × × 10 down-sampling replicates (60 runs)
Window extraction	<code>prepare_gnn_data_10.py</code> , <code>-pe_threshold 0.8</code> , <code>-window 20</code> , GRCh38 no-alt analysis set
Truth set	v5.0q benchmark, chr1–chr22 small variants
Split (by chromosome)	train chr1–chr14, chr18–chr20; validation chr15, chr16; test chr17, chr21, chr22
Windows / scored nodes	train 602,788 / 18,356,880; val 67,880 / 1,509,576; test 51,448 / 1,367,580
Class prevalence (train)	correct 17,812,710 (97.04%), error 544,170 (2.96%); 21.5% of windows contain an error node
Labels	anchor-relative within each PS; single disagreement → <i>unphase</i> if $PE \geq 0.9$ else <i>flip</i> ; run of ≥ 2 → first variant <i>break</i> , rest <i>correct</i> ; merged to one <i>error</i> class
Masked from loss	variants absent from the benchmark; all SV and 5mC records (the truth set holds small variants only); nodes outside the central zone
<i>Optimization</i>	
Loss	focal cross-entropy, $\gamma = 1.5$, label smoothing 0.05, on a per-batch subset of all error nodes plus $\leq 4\times$ as many randomly drawn correct nodes
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$, weight decay 10^{-3}
Learning rate	3×10^{-4} , no warm-up, cosine annealing to 10^{-5} over 100 epochs
Batch size	64 windows, batched as disjoint graphs
Regularization	dropout 0.4 (local branch, global branch, FFN; inactive at inference); gradient-norm clipping 1.0
Graph preprocessing	reverse edge added per DOT edge with identical features (undirected)
Augmentation (train only)	haplotype swap, $p = 0.5$; coverage down-scaling $\times U[0.4, 0.7]$, $p = 0.5$, error-bearing windows only
Epochs	budget 100, early stopping after 25 epochs without improvement; stopped at 69
Seed	42 (Python and PyTorch generators)
Environment	PyTorch 2.11.0, PyTorch Geometric 2.7.0, Python 3.12, CUDA 13.0, cuDNN 9.19, one NVIDIA GeForce RTX 5090
<i>Selection and export</i>	
Selection criterion	max over 81 thresholds in $[0.01, 0.99]$ of $R(2 - 1/P)$ subject to $R \geq 0.005$, on chr15–chr16; $P - 1$ where $P \leq 0.5$
Selected checkpoint	epoch 44; $S = 0.0235$ at $P = 0.626$, $R = 0.058$, threshold 0.868
Held-out test (chr17, 21, 22)	argmax: accuracy 0.941, error-class $P/R/F_1 = 0.258/0.499/0.340$; tuned threshold: 0.946, 0.273/0.468/0.345; $n = 41,378$ error nodes
Deployment threshold	0.30, more permissive than the selection threshold because the only action is to withhold phase
Export	ONNX opset 17, dynamic N ; base64 float32 blob compiled into <code>GNNWeights.h</code> ; C++ vs. ONNX Runtime max. absolute difference 3×10^{-7}

Supplementary Table 5: Parameters of longphase phase and their defaults.

Parameter	Default	Meaning
<code>-ont / -pb</code>	–	sequencing platform (required; <code>-ont</code> enables the homopolymer SNV filter)
<code>-indels</code>	off	phase small indels
<code>-indelQuality</code>	0	exclude indels with QUAL below this value
<code>-sv-file, -mod-file</code>	–	SV VCF (Sniffles2/cuteSV with read names); 5mC VCF from modcall
<code>-q, -mappingQuality</code>	1	minimum mapping quality of alignments
<code>-x, -mismatchRate</code>	3	mark reads as unreliable above this mismatch rate
<code>-p, -baseQuality</code>	12	base quality below which an observation is down-weighted
<code>-e, -edgeWeight</code>	0.1	weight of a down-weighted observation (otherwise 1)
<code>-a, -connectAdjacent</code>	35	number of downstream variants each variant is linked to
<code>-d, -distance</code>	300,000	a variant is skipped (no votes, no haplotype) when the next heterozygous variant is further away than this; edges and votes themselves are not distance-limited
<code>-l, -edgeThreshold</code>	0.7	pair casts no vote if $\min(P, Q) / \max(P, Q)$ exceeds this (0.3 for SNV–5mC)
<code>-L, -overlapThreshold</code>	0.2	overlap fraction above which the shorter of two alignments of a read is dropped
<code>-w, -svWindow</code>	20	CIGAR operations scanned around an SV start (the help text prints <code>-sv-window</code> , which does not parse)
<code>-h, -svThreshold</code>	0.10	relative length tolerance for a CIGAR indel to support an SV (help text: <code>-sv-threshold</code>)
<code>-m, -readConfidence</code>	0.65	minimum majority fraction to tag a read in read-based correction
<code>-n, -snpConfidence</code>	0.75	minimum confidence to keep a variant’s phase after read-based correction
<code>-dot</code>	off	export the phasing graph (required by <code>gnn</code>)

Supplementary Table 6: Parameters of longphase modcall and their defaults. Values are the compiled-in defaults (ModCall.cpp:67–73 at commit cc17fb1); the on-line help text advertises 0.7 for `-heterRatio` and 6 for `-connectAdjacent` and the long option for the BAM input is `-methylbamfile` ([fix the help text and confirm the values used for the benchmarks](#)).

Parameter	Default	Meaning
<code>-m, -modThreshold</code>	0.8	ML probability at or above which a base is methylated
<code>-u, -unModThreshold</code>	0.2	ML probability at or below which a base is unmethylated
<code>-e, -heterRatio</code>	0.6	minimum $\min(m, u) / \max(m, u)$ for a heterozygous site
<code>-i, -noiseRatio</code>	0.2	maximum fraction of reads with intermediate probability
<code>-a, -connectAdjacent</code>	20	neighbouring variants searched for SNV co-segregation
<code>-c, -connectConfidence</code>	0.9	minimum majority ratio of co-segregating reads
<code>-all</code>	off	also output homozygous modification sites

Supplementary Table 7: Parameters of longphase gnn and longphase haplotag.

Parameter	Default	Meaning
<i>gnn</i>		
<code>-s, -snp-file</code>	–	phased SNV/indel VCF from phase
<code>-sv-file, -mod-file</code>	–	phased SV and 5mC VCFs (corrected copies are written)
<code>-dot-prefix</code>	from <code>-s</code>	prefix of the DOT files
<code>-r, -reference</code>	–	reference FASTA (enables genomic-context features)
<code>-pe-threshold</code>	0.8	phasing entropy that opens a window
<code>-window</code>	20	variants on each side of the trigger
<code>-B, -break-threshold</code>	0.30	error probability at or above which a variant is unphased
<code>-respect-bridge</code>	off	never unphase bridge vertices
<code>-no-split-blocks</code>	off	keep the original PS even if the block becomes disconnected
<i>haplotag</i>		
<code>-q, -qualityThreshold</code>	1	minimum mapping quality of tagged alignments
<code>-p, -percentageThreshold</code>	0.6	minimum fraction of a read’s informative alleles on the majority haplotype
<code>-tagSupplementary</code>	off	also tag supplementary alignments
<code>-sv-file, -mod-file</code>	–	use phased SVs and 5mC sites as additional tagging evidence
<code>-cram</code>	off	write CRAM instead of BAM

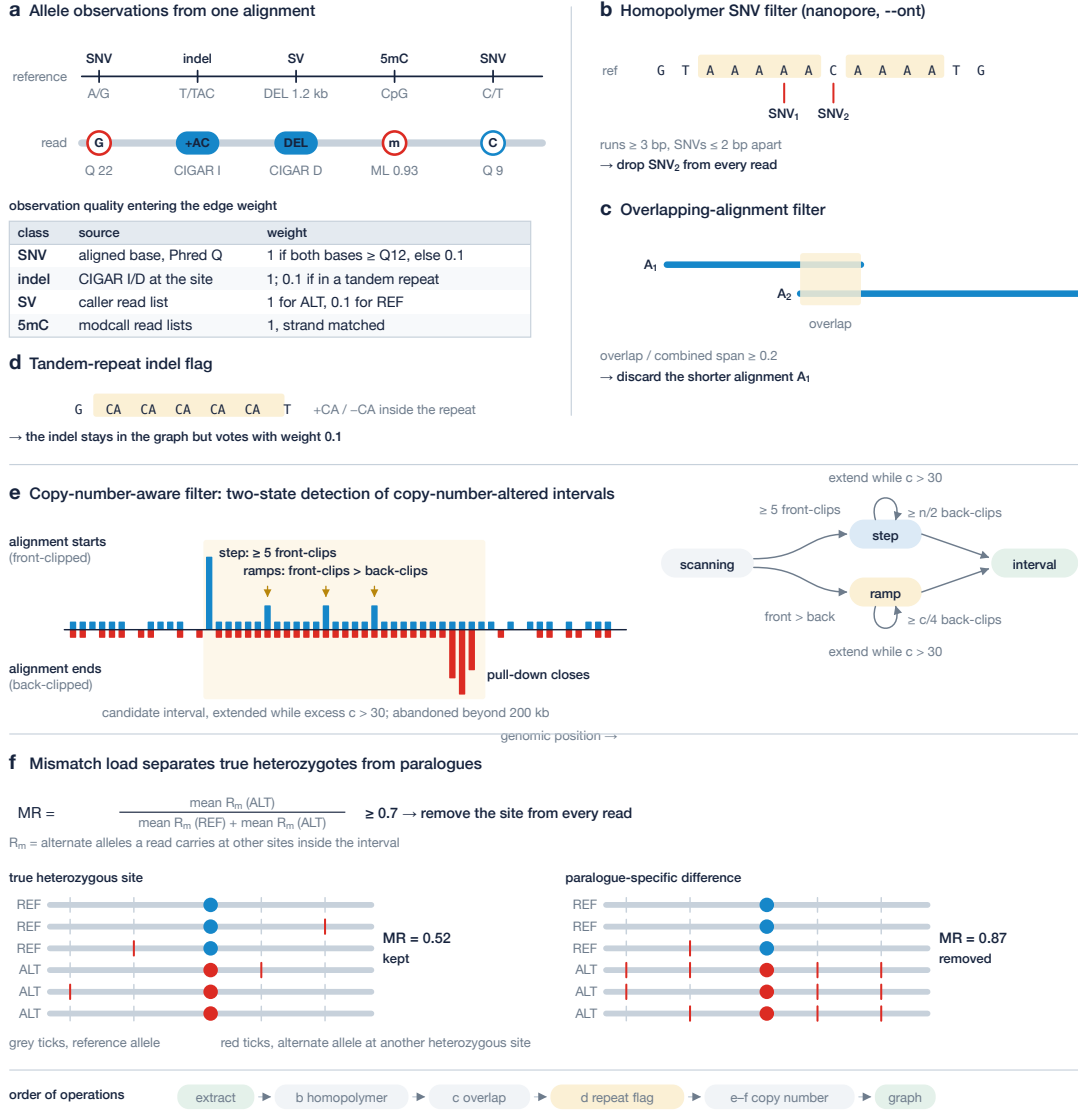
Supplementary Table 8: Software and data used in the benchmarks. [Complete versions and accessions.]

Item	Version	Notes
HG002 nanopore R10.4.1 reads	[–]	[flow cell, kit, basecaller model, source]
Reference	GRCh38 [build]	
minimap2	[–]	-ax map-ont -y
Clair3	[–]	SNV and indel calls
Sniffles2	[–]	-output-rnames
GIAB v4.2.1 HG002	v4.2.1	GRCh38-based small-variant benchmark
T2T-HG002 v1.1 small-variant benchmark ^{1,2}	GIAB v5 draft	projected onto GRCh38; referred to as v5.0q
WhatsHap	2.8	[options]
HapCUT2	1.3.4	default parameters
LongPhase 1.0	1.0.x [exact]	-ont
LongPhase 2	[release]	-ont; gnn with defaults

Supplementary Table 9: Selected published tools, pipelines and studies that use LongPhase.
 Compiled from Europe PMC (80 indexed citing articles) and OpenAlex (96 citing works) for 10.1093/bioinformatics/btac058, September 2026; the “use” column quotes or paraphrases the citing article’s Methods.

Work	Journal, year	Use of LongPhase
<i>Variant callers and pipelines</i>		
ClairS ³	Nat. Methods 2026	default germline phasing of tumour and normal alignments (<code>-whatshap_for_phasing</code> the alternative); haplotagging selectable
ClairS-TO ⁴	Nat. Commun. 2025	default intermediate phasing and haplotagging of the tumour alignment (WhatsHap selectable)
Clair3-RNA ⁵	Nat. Commun. 2025	phasing in the long-read RNA pipeline
Clair3 (accelerated) ⁶	Bioinformatics 2026	selectable intermediate phasing and haplotagging (<code>-longphase_for_phasing</code> ; WhatsHap the default)
DeepVariant local haplotagging ⁷	Nat. Commun. 2024	not used; cited as one of the haplotaggers (with WhatsHap and Margin) that Clair3 and Medaka employ
Severus ⁸	Nat. Biotechnol. 2025	one of the documented routes to the haplotagged BAMs it consumes (WhatsHap the other)
EPI2ME wf-human-variation ⁹	ONT workflow	intermediate per-contig phasing (<code>phase_contig</code>); WhatsHap for final phasing and haplotagging
HiFi-somatic-WDL ¹⁰ cuteHap ¹¹	PacBio workflow Adv. Sci. 2026	optional phasing tool read phasing before haplotype-aware SV calling
SpecImmune ¹²	Adv. Sci. 2026	joint SNV–SV phasing of immune genes
<i>Biological and clinical studies</i>		
Fluctuating DNA methylation in cancer ¹³	Nature 2025	read phasing (v1.7) and <code>modcall</code> for CpG methylation
Advanced cancer cohort (POG) ¹⁴	Cell Genomics 2024	discussed as phasing method for tumours
Pediatric epilepsy episignatures ¹⁵	Nat. Commun. 2024	phasing of Clair3 and SV calls
GNAS transcripts in PHP1B ¹⁶ 3-hour genome for neonatal risk ¹⁷	PNAS 2025 Genet. Med. Open 2024	phasing of Clair3 calls real-time phasing during sequencing
Ultrarapid diagnostics in critical care ¹⁸	Eur. J. Hum. Genet. 2025	read phasing (v1.0.7)
HYDIN in primary ciliary dyskinesia ¹⁹	Eur. J. Hum. Genet. 2024	phasing of Clair3 calls
Antisense oligonucleotides in APBD ²⁰	Nucleic Acids Res. 2025	phasing of Clair3 calls
Native CpG methylation fine-mapping ²¹	HGG Adv. 2026	phasing and haplotagging (chosen for speed and SV support)
Canadian national platform benchmark ²²	Front. Genet. 2026	phasing inside the Clair3 workflow
Pallas’s cat genome ²³	NAR Genom. Bioinform. 2023	haplotagging for allele-specific methylation
<i>Methods evaluated against LongPhase</i>		
HiPhase ²⁴	Bioinformatics 2024	comparison method
Ralphi ²⁵	Genome Res. 2025	comparison method (fastest runtime on ONT/PacBio)
GCphase ²⁶	BMC	comparison method

Supplementary Figures

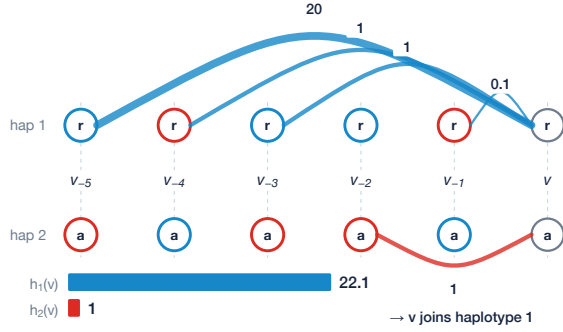


Supplementary Fig. 1: Read-level allele observations and pre-graph filters. **a**, For every alignment, each overlapped variant yields an allele observation with an associated quality: SNVs from the aligned base (Phred quality), indels from CIGAR operations starting at the site, SVs from CIGAR operations of matching length or from the caller's read list, and 5mC from the `modcall` read lists. The table summarizes how each class enters the edge weights. **b**, Homopolymer SNV filter, applied under `--ont`: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats are flagged and vote with weight 0.1. **e**, Copy-number-aware filter, interval detection. Per-position counts of alignments starting (front-clipped, above the axis) and ending (back-clipped, below) are scanned by a two-state detector (right): a *step* boundary (≥ 5 front-clips at one position) marks a copy-number transition; a *ramp* boundary (front-clips exceeding back-clips without reaching that threshold) marks the nested breakpoints left by breakage–fusion–bridge cycles. Both states carry a running excess c of front- over back-clips and extend the candidate end by a 30 kb look-ahead while $c > 30$; a step closes at an adaptive pull-down threshold ($n/2$ for an opening front-clip count $n \geq 10$, else 5) or when c decays to its floor, a ramp closes when back-clips reach $c/4$. Candidates longer than 200 kb are abandoned. **f**, Copy-number-aware filter, site removal. Inside an interval, each read is scored by its mismatch load R_m , the number of heterozygous sites in the interval at which it carries the alternate allele, and each heterozygous site by $MR = \bar{R}_m^{\text{ALT}} / (\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}})$. At a genuine heterozygous site both allele populations are ordinary reads of the locus and $MR \approx 0.5$; at a paralogue-specific difference the alternate allele is carried by reads that originate elsewhere and mismatch the reference at many other sites in the same interval, so $MR \rightarrow 1$. Sites with $MR \geq 0.7$ are removed from all reads; sites at which either mean is zero are never removed. The strip at the bottom gives the order in which the filters are applied; the copy-number-aware filter (**e,f**) acts last, immediately before graph construction.

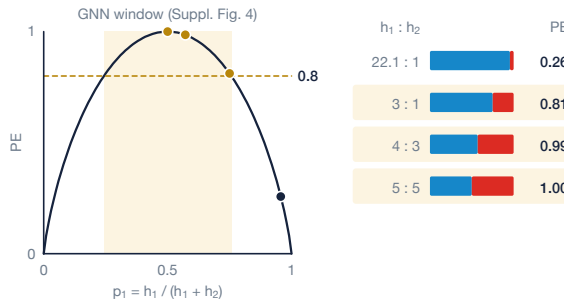
a Allele-pair support between u and v



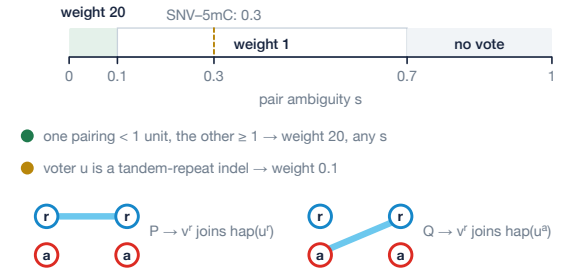
c Weighted votes from up to $k = 35$ upstream variants



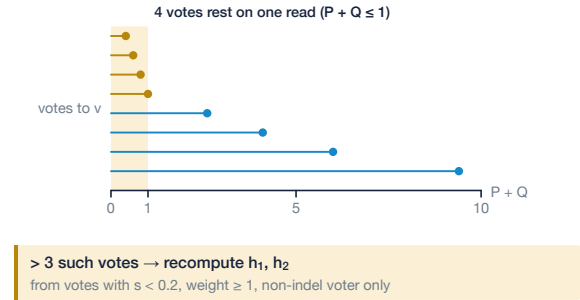
e Phasing entropy $PE = -\sum p_i \log_2 p_i$



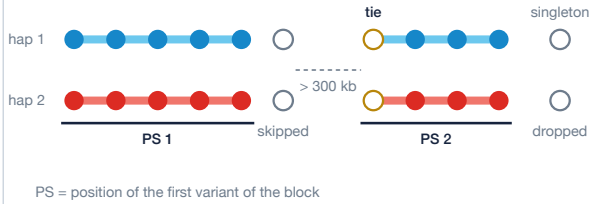
b Vote rule for the pair (u , v)



d Single-read guard



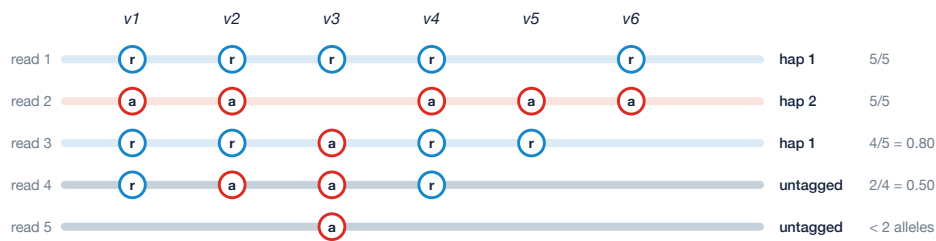
f Block formation



Supplementary Fig. 2: Pairwise allele support, weighted voting and phasing entropy. a,

Between two variants u and v the reads accumulate four weights ($w_{rr}, w_{ra}, w_{ar}, w_{aa}$), each read adding 1 when both bases have quality $\geq Q_{12}$ and 0.1 otherwise. $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$ are the supports for the two possible pairings and $s = \min(P, Q) / \max(P, Q)$ their ambiguity. **b,** The pair votes for the better-supported pairing: at weight 20 when $s \leq 0.1$ with $P + Q \geq 1$ or, irrespective of s , when one pairing has less than 1 unit of support and the other at least 1; at weight 1 otherwise; and not at all when $s > 0.7$ (0.3 for SNV-5mC pairs, which also cast no vote when $P + Q < 1$). A voting variant that is a tandem-repeat indel votes at weight 0.1. The vote names a haplotype for v relative to the haplotype already assigned to u . **c,** Variant v receives weighted votes from up to $k = 35$ upstream variants that share a read with it, at any distance; line thickness encodes vote weight. The weighted totals h_1 and h_2 decide the haplotype; a tie opens a new phase block. **d,** Guard against a single erroneous ultra-long read: among the votes received by v , those resting on a single read ($P + Q \leq 1$, amber) are counted, and if more than three exist, h_1 and h_2 are recomputed from the votes with $s < 0.2$, weight ≥ 1 and a non-indel voter only. **e,** Phasing entropy of the vote totals, $PE = -\sum_i p_i \log_2 p_i$ with $p_i = h_i / (h_1 + h_2)$, as a function of the vote split (left) and for four examples (right); values ≥ 0.8 (shaded) open a window for the neural network (Supplementary Fig. 4). PE, h_1 and h_2 are written to the VCF as INFO/PE, INFO/H1 and INFO/H2. **f,** Block formation. Variants are visited in coordinate order; a variant whose next heterozygous variant lies more than 300 kb away ($-\text{distance}$) is skipped and casts no votes, a variant that receives no informative vote or a tie (amber) starts a new phase set, and blocks holding a single variant are dropped (grey). Haplotype labels propagate along the block as the phased genotypes 0|1 and 1|0; the phase set (PS) is the position of its first variant.

a Step 1 — assign each read to a haplotype

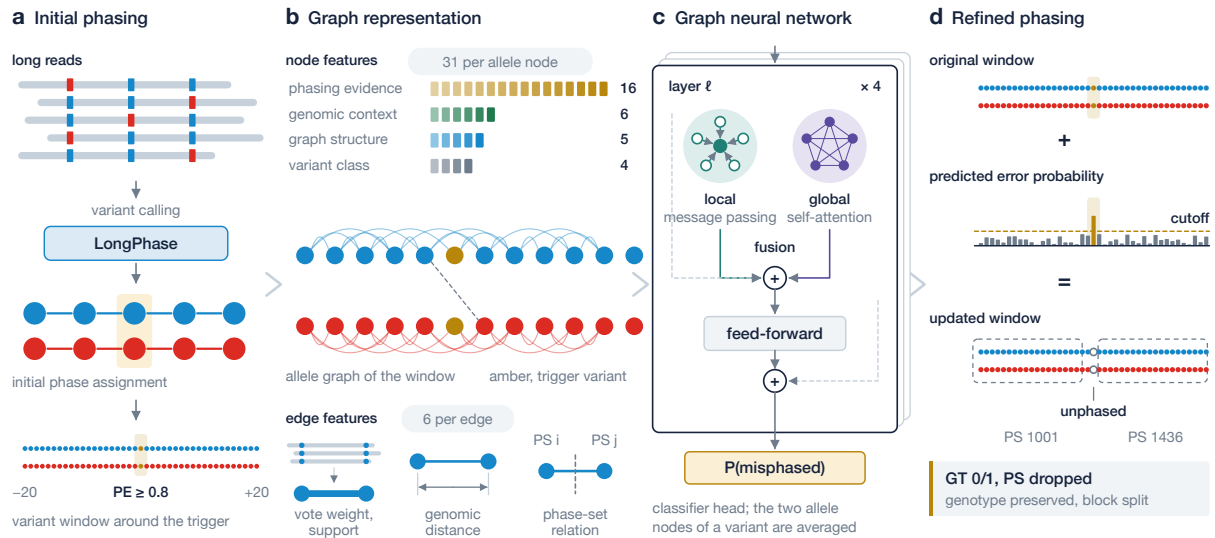


tagged if $\max(n_1, n_2) / (n_1 + n_2) > 0.65$ and weighted evidence > 1
 allele weights: SNV 1, SV 1, indel 0.1, 5mC 0 (so ≥ 2 SNV/SV alleles are needed)

b Step 2 — re-derive each variant's phase from tagged reads

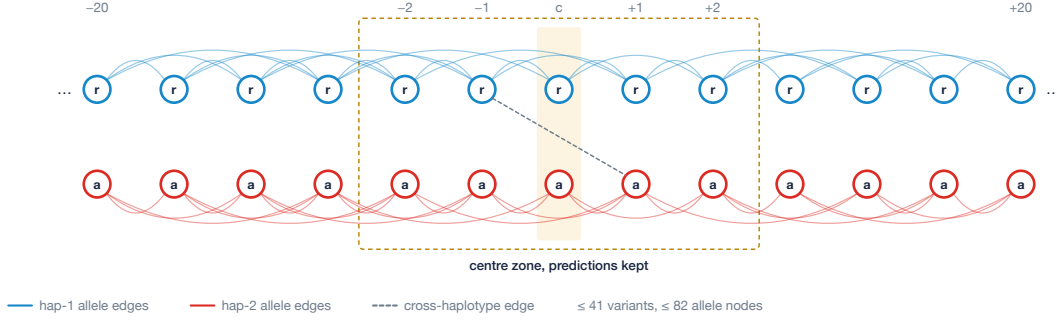


Supplementary Fig. 3: Read-based correction. **a**, Each read is assigned to a haplotype by majority over its phased SNV and SV alleles (indels count 0.1, 5mC 0); reads with majority ≤ 0.65 or fewer than two informative alleles remain untagged. **b**, Each variant's phase is re-derived from the tagged reads and retained only if the consistent fraction ρ exceeds 0.75; otherwise the variant is unphased. Only variants supported by both read populations keep a phase, so assignments driven by a few chimeric or mis-mapped reads are removed. The two thresholds are `-readConfidence 0.65` and `-snpConfidence 0.75`.

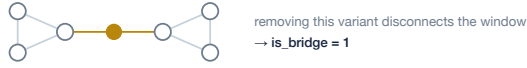


Supplementary Fig. 4: Overview of the graph neural network correction module. **a**, LongPhase phases the called variants from the long-read alignments; every phased variant whose phasing entropy reaches 0.8 (amber) becomes the trigger of a window of ± 20 variants. **b**, The window is encoded as a graph whose nodes are the two alleles of each variant and whose edges are the surviving DOT edges (Supplementary Fig. 5d). Each allele node carries 31 features in four groups and each edge six features (Supplementary Tables 1 and 2); the shaded squares stand for the individual features of a group, not for their values. **c**, Each of four GPS layers passes the graph through a local branch, in which a node attends to its neighbours in the phasing graph (GATv2), and a global branch, in which every allele node in the window attends to every other. The two representations are fused additively with a skip connection from the layer input (dashed grey) and passed through a feed-forward block with a second skip connection. A classifier head then assigns each allele node a probability of being misphased, and the two allele nodes of a variant are averaged. **d**, Variants whose predicted error probability reaches 0.30 are unphased: the genotype is written unphased and without a phase set, the genotype itself is never flipped, and connectivity over the surviving edges splits the phase set in two. Detailed views of the window, the network and the phase-set update are given in Supplementary Figs. 5 and 6.

a A window is opened around every phased variant with $PE \geq 0.8$



b Bridge-vertex feature



c Centre-zone mask

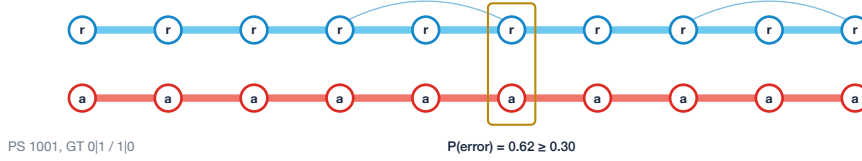


d Exported phasing graph (DOT), the input of the module

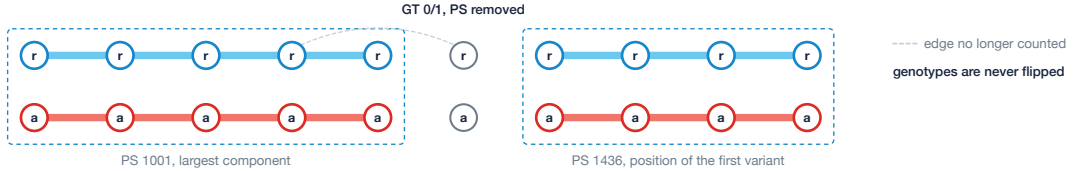
1001.1 → 1543.1 [label=16.0] 1001.2 → 1543.2 [label=1.1]

position.allele → position.allele, label = vote weight cast at the source

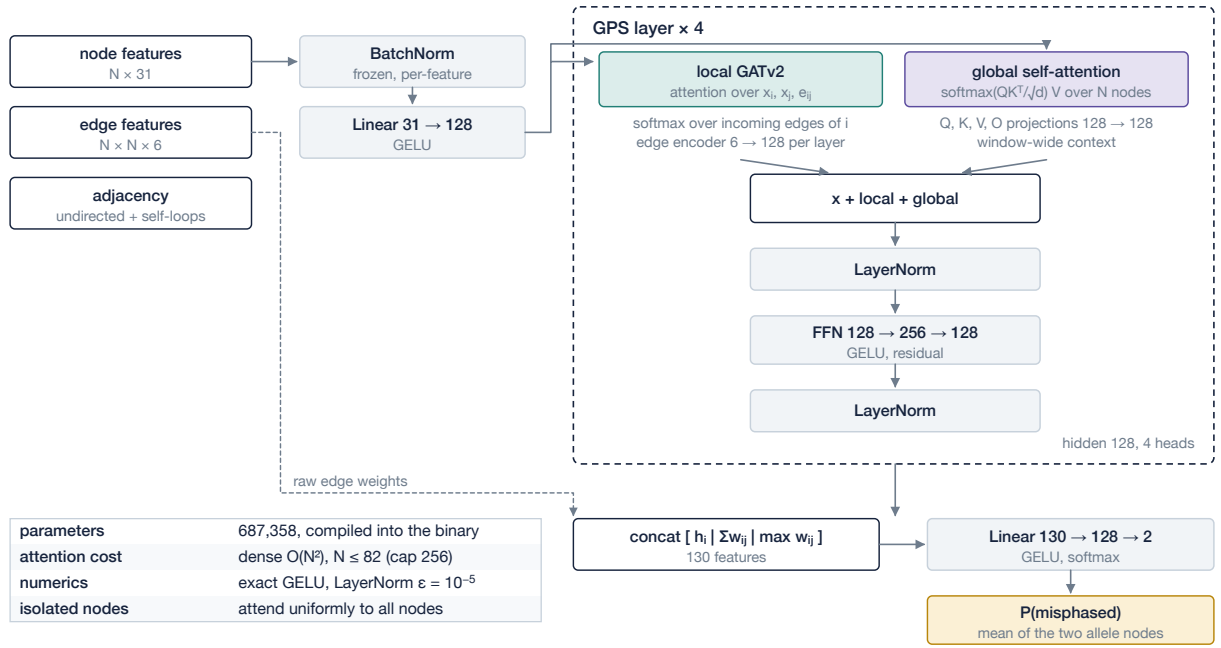
e Before correction — one phase set



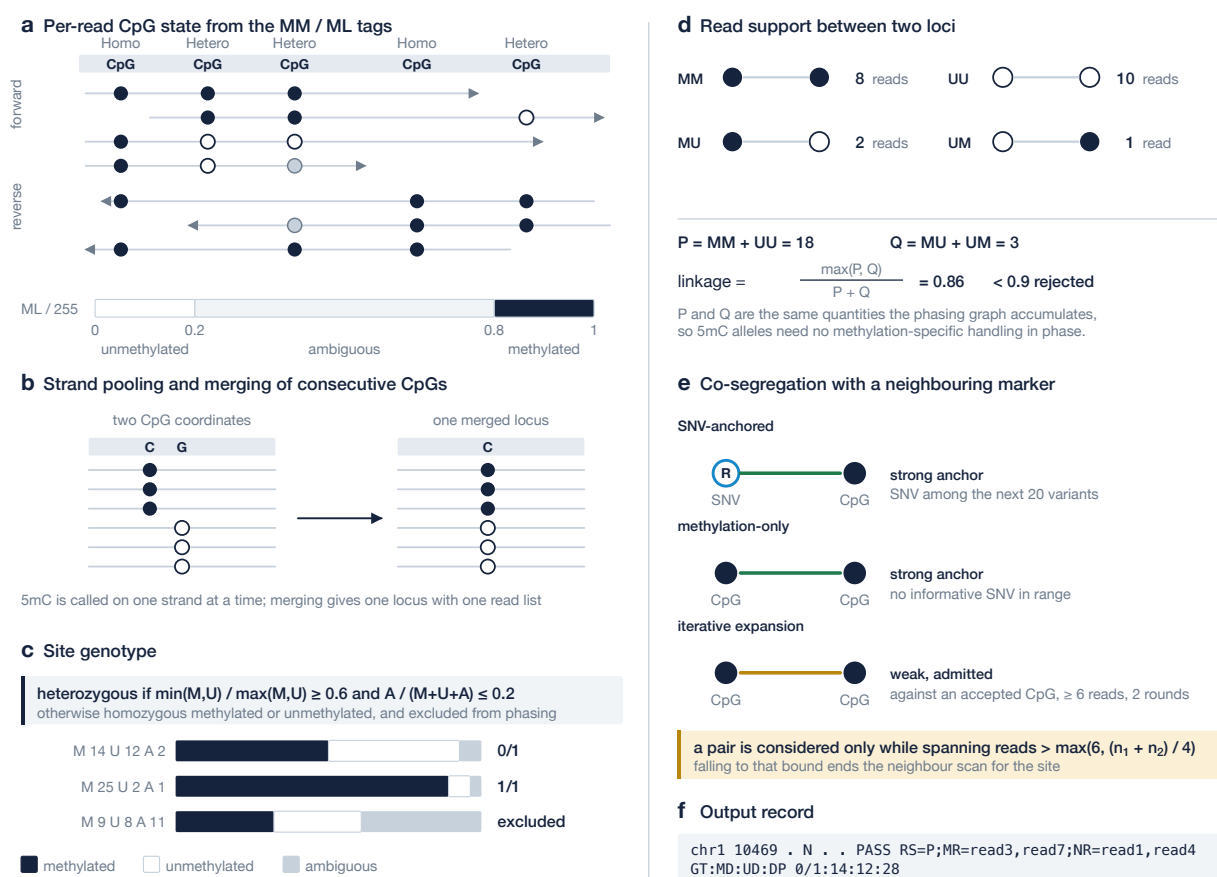
f After correction — variant unphased, phase set split



Supplementary Fig. 5: Window construction and phase-set update for the graph neural network. **a**, A window of ± 20 variants (two allele nodes each) is opened around every phased variant with $PE \geq 0.8$; edges are the DOT edges with both endpoints in the window (blue and red arcs) plus self-loops; a dashed grey edge marks a cross-haplotype link. Predictions are read only in the central zone. **b**, Bridge-vertex feature: a variant whose removal disconnects the window graph. **c**, Definition of the central zone. A window holds at most 41 variants and 82 allele nodes (a 256-node cap exists but is never reached at the default window size); SNV, indel, SV and 5mC records are merged and sorted by position before nodes are created. Because offsets are normalized by the largest offset in the window, $r_{\max} = 1$ in practice and the retained zone is the central half of the window's genomic span. The same rule, with the same constant, defines which labelled nodes carried a training loss, so the zone scored at inference is the zone the network was trained on. Edges are the DOT edges with both endpoints in the window, symmetrized, plus one self-loop per node whose features are the mean over that node's incident edges. The predictions of the two allele nodes of a variant are averaged, and a variant covered by several windows receives the mean over those windows. **d**, The exported phasing graph (DOT) that the module reads: one line per voting pair, source and target written as position.allele, and the vote weight accumulated at the source as the label. **e**, Phase-set update, before correction: a variant with predicted error probability ≥ 0.30 (amber) inside a phase set. **f**, After correction the variant is written as an unphased genotype without PS; connectivity of the remaining variants is recomputed over the surviving DOT edges, the largest component keeps the original PS and every other component receives a new PS equal to the position of its first variant. Genotypes are never flipped.



Supplementary Fig. 6: Graph neural network architecture. Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph (edge features enter through a per-layer edge encoder) with a global self-attention branch over all nodes, fused additively into the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and the non-negative maximum, over the node's incoming edges excluding its self-loop, of the logistic-transformed edge weight $1/(1 + e^{-w})$, and classified into phased-correctly versus misphased; the probabilities of the two allele nodes of a variant are averaged.



Supplementary Fig. 7: Allele-specific methylation calling with modcall. **a**, Per-read CpG state is read from the MM/ML tags on both strands: methylated at $ML/255 \geq 0.8$, unmethylated at ≤ 0.2 , ambiguous in between. Only CpG dinucleotides are considered. **b**, 5mC is called on one strand at a time, so reverse-strand calls are mapped to the forward CpG coordinate and runs of consecutive CpGs are merged into a single locus carrying one read list. This removes strand from the problem and lets phase vote on methylation loci with the unmodified graph algorithm. **c**, A site is heterozygous when $\min(M, U) / \max(M, U) \geq 0.6$ and $A / (M + U + A) \leq 0.2$, for methylated, unmethylated and ambiguous read counts M , U and A ; otherwise it is homozygous and excluded. **d**, Balance alone does not imply allele specificity, since stochastic methylation satisfies **c** as well. Read support between two loci is therefore decomposed into the four state pairs, whose concordant and discordant totals are exactly the quantities P and Q that the phasing graph accumulates (`VariantEdge::findNumberOfRead`); a pair is linked when $\max(P, Q) / (P + Q) \geq 0.9$. The example shown falls below the threshold and is rejected. **e**, Linking runs in three modes: against an SNV among the next 20 variants, against another candidate CpG where no informative SNV is in range, and an iterative expansion that admits weak candidates against an already accepted CpG. In every mode a pair is considered only while the reads spanning it exceed $\max(6, (n_1 + n_2) / 4)$, and falling to that bound ends the neighbour scan for that site rather than skipping one pair. **f**, Output record; the MR and NR read names attach the 5mC alleles to reads in phase. [Two real-data panels would close the remaining gaps, and both exist in the group's thesis material. (i) Per-site ML probability at heterozygous versus homozygous CpGs with the 0.8 and 0.2 lines marked, showing that it is the homozygous sites that spread across the ambiguous band; the original was computed on 48 sites in the imprinted *SNRPN* locus, too small and too special a sample for a Nature Methods supplement, so recompute genome-wide on HG002. (ii) Distribution of the linkage ratio in **d** stratified by the true class of the site pair (heterozygous–heterozygous, homozygous–homozygous, heterozygous–homozygous). This is the strongest justification available for the 0.9 threshold: in the original the three classes separate at roughly 0.65–1, 0.5–0.75 and 0.5–0.7, showing that the cut removes homozygous sites misclassified as allele-specific rather than being an arbitrary round number. Regenerate both with the submitted version.]

a Switch errors, flips and Hamming distance within one block



b Block statistics



Supplementary Fig. 8: Phasing evaluation metrics. **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Block intersection and the block-level statistics. Only variants that are heterozygous in both files and phased in both are compared (dark bars); blocks left with a single variant are ignored (outline). Phased fraction, block N50 and switch error rate are computed over the intersected blocks as implemented in `longphase compare`.

Placeholder: panels to be regenerated with the submitted version.

Supplementary Fig. 9: Calibration of the voting and read-filtering constants. Each panel is a sweep on HG002 at 10–60 $\times$; the value adopted in LongPhase 2 is marked. **a**, Distribution of pair ambiguity $s = \min(P, Q) / \max(P, Q)$ over all assessed pairs and over the subset that participate in a switch error, showing that both mass and residual error concentrate below $s = 0.1$ and motivating $\alpha = 0.1$ as the boundary of the up-weighted class. **b**, Switch errors and block N50 as a function of the no-vote threshold β . Switch errors fall monotonically as β decreases, so switch error alone does not identify an optimum; $\beta = 0.7$ is the smallest value at which block N50 is not yet degraded, and is adopted on that basis. **c**, Switch errors as a function of the up-weight w applied when $s < \alpha$, at each coverage; the reduction saturates by $w \approx 20$. **d**, Distribution of base quality over all bases and over bases carrying a sequencing error, showing errors concentrated below Q10 and motivating $\theta = 12$; and switch errors as a function of θ . **e**, Switch errors over the grid of single-read-guard parameters N (number of single-read votes that triggers the guard) and μ (ambiguity cut applied to the surviving votes), and switch errors with and without the guard at each coverage. [All five panels exist as figures in the group's thesis material (Po-Hui Tzeng, 2026, Figs. 3.7, 3.10–3.12, 3.14, 3.17, 3.18, 3.21) and should be regenerated with the submitted version on the Clair3 calls and coverages used here, then redrawn to a single consistent style. Two caveats to carry over honestly: in **b** the switch-error curve does not have an interior optimum, so the choice of β rests on the N50 trade-off and must be presented that way; and in **e** the guard slightly *increased* switch errors at 10 $\times$ in the original sweep, which should be shown rather than cropped.]

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